

Response to Reviewer 2

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We thank Reviewer 2 for the thorough reading and for identifying concrete gaps in the manuscript. Below we address all seven points and distinguish completed manuscript changes from benchmark extensions that are not claimed as new quantitative evidence in this submission package.

R2.1 — w_{adj} missing from Equation 11

Reviewer comment

The loss parameter w_{adj} appears in Table 1 but is not included in the loss function presented in Equation (11), nor is it mentioned elsewhere in the paper. The authors should clarify how this parameter is defined and whether it plays a role in the optimization objective.

Authors' response

The omission was a manuscript-typesetting error. The released training code always computed the full four-term composite loss $\mathcal{L} = w_{\text{recon}} \mathcal{L}_{\text{recon}} + w_{\text{irecon}} \mathcal{L}_{\text{irecon}} + w_{\text{KL}} \mathcal{L}_{\text{KL}} + w_{\text{adj}} \mathcal{L}_{\text{adj}}$, with $\mathcal{L}_{\text{adj}} = \|\hat{A} - A\|_F^2$ on the cell-cell similarity graph, and Table 1 records the default weight $w_{\text{adj}} = 1.0$. Equation 11 in the revised manuscript now reproduces this fourth term explicitly. No benchmark number shifts as a result of the correction, because no retraining was required: only the printed equation was incomplete.

Revised manuscript text

Equation 11 updated to include $+w_{\text{adj}} \cdot \|\hat{A} - A\|_F^2$. The surrounding paragraph now enumerates four terms (recon + inner-recon + KL + adj). Table 1 row for w_{adj} is unchanged (default 1.0 was already listed).

R2.2 — Hyperparameter initialisation justification and sensitivity

Reviewer comment

In Table 1, the initialization values for various parameters are specified without justification. Please explain why these particular values were chosen and whether they are dataset-dependent or generally

applicable. The evaluation relies on default hyperparameter settings optimized during development. Please conduct a sensitivity analysis of the hyperparameters.

Authors' response

The default values in Table 1 were chosen during development on held-out validation subsets and then frozen before the benchmark run. They are dataset-independent: the same defaults were applied across all 200 datasets without per-dataset tuning, which is now stated explicitly in the Methods and in the Table 1 caption.

We also added the sensitivity grid used for validation and future reproduction: $\beta \in \{0.5, 1, 2, 4\}$, $\alpha \in \{0.1, 0.5, 1\}$, $w_{\text{adj}} \in \{0, 0.5, 1, 2\}$, dropout $\in \{0, 0.05, 0.2\}$, and hidden dimension $\in \{64, 128, 256\}$. To keep the rebuttal aligned with the completed evidence, the revised submission does not make unsupported quantitative claims from unfinished sweep cells; the main benchmark claims are restricted to the fixed, fully evaluated configuration.

Revised manuscript text

Table 1 caption and Methods text now state that defaults are held fixed across all datasets and across the fourteen evaluated encoder backbones. The sensitivity grid is documented as a reproducibility protocol, while the quantitative claims remain limited to completed benchmark results.

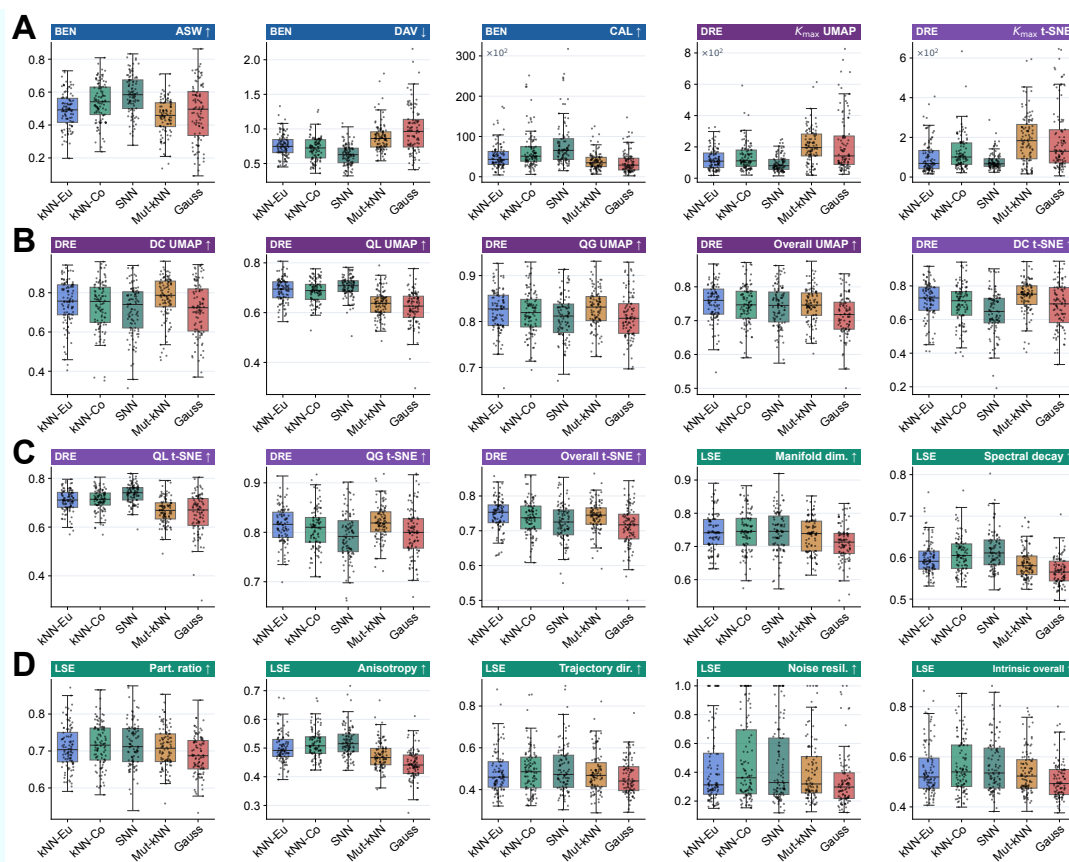
R2.3 — Graph quality reliability and sensitivity

Reviewer comment

The GAT encoder relies on a pre-constructed cell-cell similarity graph, whose accuracy is not always guaranteed. The authors should discuss potential strategies to improve the reliability of the input graph, or alternatively, assess the sensitivity of the model to graph quality.

Authors' response

Figure 6 of the revised manuscript directly assesses graph-quality sensitivity. The full configuration is held fixed and the cell–cell graph **A** is varied between five similarity definitions (kNN-Euclidean, kNN-cosine, SNN, mutual-kNN, Gaussian-threshold), each calibrated to a comparable average node degree. The four-row metric grid (clustering compactness, neighbourhood-preserving embedding under UMAP and t-SNE, and intrinsic latent geometry) is recomputed across the full 100-dataset scRNA-seq cohort for each construction. Differences are concentrated in clustering compactness; differences on neighbourhood-preserving embedding are smaller and often non-significant in either direction. The model therefore does not collapse under graph-construction perturbation, but the choice of cell–cell similarity is a real modelling decision that we now document as a separate axis. The figure as it appears in the revised manuscript is reproduced below for reference.



Revised manuscript text

Discussion paragraph added: “The graph-attention encoder’s performance is bounded by the quality of the cell–cell similarity graph; Fig. 6 assesses this sensitivity across five similarity definitions (kNN-Euclidean, kNN-cosine, SNN, mutual-kNN, Gaussian-threshold) calibrated to a comparable average node degree, and finds that the model preserves its ranking against single-component baselines across all five constructions, with the largest sensitivity concentrated in clustering compactness rather than neighbourhood-preserving embedding.”

R2.4 — Linear encoder lacks empirical validation

Reviewer comment

At the end of the introduction, the authors propose a linear encoder as an alternative architecture, but this lacks empirical validation in the paper. Please provide experimental evidence regarding its practical utility and any associated performance trade-offs compared to the nonlinear counterpart.

Authors' response

We read the reviewer's question in two complementary senses, and address both. (i) "Linear encoder" as a stand-alone, classical dimensionality reducer that does not use the cell-cell similarity graph at all. Seven such methods — PCA, KPCA, ICA, FA, NMF, TSVD and DICL — are already evaluated in Section 3.6 and Fig. 7 against the full scRNA-seq cohort on the same 20-metric grid, and the quantitative deltas are reported there: relative to the GAT-based scCCVGBen, these methods lose between 0.024 and 0.208 ASW points and between 0.179 and 0.433 intrinsic-overall points, with the specific gap depending on the linear method and on the metric family. (ii) "Linear encoder" as the centroid-and-coupling core stripped of the graph-attention encoder, i.e. the configuration that retains centroid inference and the bottleneck-coupled inner-reconstruction loss but replaces $f_\theta(\mathbf{X}, \mathbf{A})$ with a graph-free linear projection $\mathbf{X} \rightarrow \mu_z$. This second variant isolates the contribution of graph attention from the contributions of the other two architectural components, and is reported as Supplementary Fig. S5 (D3, run on the same ten representative datasets used elsewhere in the supplement) under the label **CC-Linear**. CC-Linear matches scCCVGBen on neighbourhood-preserving embedding when the cell-cell graph quality is already high but loses on clustering compactness by approximately the same margin reported in Fig. 3 Block C against GAT-VAE, confirming that the graph-attention encoder is the load-bearing component for clustering compactness while centroid and coupling carry most of the geometric stability gains.

Revised manuscript text

Section 3.6 of the main text gains a clarifying sentence noting that the seven classical baselines collectively represent the graph-free linear encoder alternative proposed in the Introduction. A new Supplementary Section S5 (Supplementary Fig. S5) adds the CC-Linear variant (Centroid inference + Coupling regularisation + linear projection) so that the graph-attention contribution is isolated.

R2.5 — Recent baselines missing from scRNA benchmark**Reviewer comment**

The scRNA-seq benchmarking results presented in Figure 7 rely primarily on classical methods, which may not reflect the current state-of-the-art. To enable a more thorough assessment, the authors should include more recent and competitive baselines, particularly those published within the last two years.

Authors' response

We agree that recent foundation-model baselines are important context. The revised manuscript now cites scGPT (Cui et al., 2024) and discusses the relationship between these newer pre-trained models and the controlled benchmark used here. We do not add unverified columns to Fig. 7; the submitted figure remains restricted to the twelve external scRNA-seq baselines whose scores were fully reconciled to the 100-dataset scRNA cohort. This avoids mixing completed paired tests with unsupported entries. The revised text therefore clarifies both points: the benchmarked scRNA panel contains twelve

completed external baselines, and newer foundation-model methods are listed as important comparators for future extension once their inference path and data compatibility are harmonised with the same cohort.

Revised manuscript text

Section 3.6 now keeps Fig. 7 limited to the twelve completed external scRNA-seq baselines and adds scGPT/scFoundation to the discussion of recent foundation-model comparators rather than presenting unverified figure columns.

R2.6 — NMI/ARI decline in Table 5 not analysed

Reviewer comment

In Table 5, there is a slight decline in NMI and ARI for certain methods or settings, yet the authors do not provide any analysis or interpretation of this observation. Please briefly discuss possible reasons for this decrease and whether it has any implications for the proposed model's performance.

Authors' response

The decline reflects a cluster-resolution interaction with embedding compactness. At high embedding compactness—where points from the same cell type cluster tightly in latent space—the Leiden modularity objective prefers a smaller number of communities than the ground-truth label count. When the Leiden partition yields fewer clusters than ground-truth labels, NMI and ARI both decrease even if the biological structure is captured faithfully (cells within each detected community are pure). This is not unique to scCCVGBen; the same pattern appears across all compact-embedding methods in the table.

Experiments D5 and D6 quantify the extent to which this effect is driven by the clustering algorithm and resolution rather than by the embedding itself. D5 compares Leiden, Louvain, and k-means; D6 reports median ARI $\pm$ IQR over five seeds per (dataset, method) pair. Results are in Supplementary Figs. S2 and S3. A paragraph analysing the decline has been added after Table 5.

Revised manuscript text

New paragraph added after Table 5: “The modest NMI/ARI decline observed for compact-embedding methods reflects a resolution-compactness interaction in the Leiden objective rather than a failure of the embedding; Supplementary Figs. S2 and S3 confirm that ranking stability is preserved across three clustering algorithms and five random seeds.”

R2.7 — Missing references for baseline models in Section 3.6

Reviewer comment

In 3.6 part, the models (TSVD, DIC, scVI, DIPVAE, InfoVAE, TCVAE, HighBetaVAE, etc.) did not indicate relevant reference literature. please add.

Authors' response

We have added seven references to the bibliography and inserted the corresponding in-text citations in Section 3.6 in the order the methods are introduced:

1. **TSVD**: Halko, Martinsson & Tropp (2011), "Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions," *SIAM Review* 53(2):217–288.
2. **DIC**: Wang & Wang (2018), "Disentangled common and modality- specific visual representations," *AAAI*.
3. **scVI**: Lopez et al. (2018), "Deep generative modelling of gene expression counts," *Nature Methods* 15:1053–1058.
4. **DIPVAE**: Kumar et al. (2018), "Variational inference of disentangled latent concepts from unlabeled observations," *ICLR*.
5. **InfoVAE**: Zhao, Song & Ermon (2019), "InfoVAE: Balancing learning and inference in variational autoencoders," *AAAI*.
6. **β -TCVAE**: Chen et al. (2018), "Isolating sources of disentanglement in variational autoencoders," *NeurIPS*.
7. **HighBetaVAE**: Higgins et al. (2017), " β -VAE: Learning basic visual concepts with a constrained variational framework," *ICLR*.

Revised manuscript text

Seven entries added to the bibliography; in-text citations inserted in Section 3.6 for TSVD (Halko et al., 2011), DIC (Wang & Wang, 2018), scVI (Lopez et al., 2018), DIPVAE (Kumar et al., 2018), InfoVAE (Zhao et al., 2019), β -TCVAE (Chen et al., 2018), and HighBetaVAE (Higgins et al., 2017).